NYMEX NATURAL GAS FUTURES (NG)

Spread & Curve Structure — Fundamental Reference

Henry Hub, Storage, Weather, Supply and the Forward Curve

This document presents a fact-based reference on the fundamental drivers of NYMEX Natural Gas futures (ticker: NG), contract size 10,000 MMBtu, with particular focus on the forces that shape the forward curve and spread structure between contract months. Natural gas is the most weather-sensitive of the major commodity futures markets: demand can swing by 30–40% between a mild and a severe winter week, while supply adjusts on a timescale of months. This mismatch between fast-moving demand and slow-moving supply makes storage the central balancing mechanism, and storage inventory data — reported weekly by the EIA — is the primary weekly fundamental signal for prompt spreads. At the same time, the shale revolution transformed the US from a supply-constrained to a supply-abundant market after 2008, and the emergence of LNG exports has re-connected the historically isolated US market to global natural gas prices. Understanding all these dimensions is the foundation of spread and butterfly trading on the NG curve.

Futures First — Internal Education Material
For Analyst Group Use

1. Contract Specifications and Delivery Mechanics

NYMEX Natural Gas futures are physically delivered contracts. The delivery point — Henry Hub, Louisiana — is the most important single physical location in the US natural gas market, the intersection of 16 interstate and intrastate natural gas pipeline systems connecting supply regions to demand centres across North America.

Parameter	Detail
Exchange	CME Group — NYMEX
Ticker	NG (front month)
Contract size	10,000 MMBtu (million British thermal units)
Quotation	USD per MMBtu; minimum tick = $0.001/MMBtu = $10.00 per contract
Contract months	All 12 calendar months; active contracts extend approximately 12 years forward (144 monthly contracts). Liquid trading concentrated in front 12–24 months.
Settlement type	Physical delivery at Henry Hub, Louisiana via pipeline at a uniform rate equal to the contract quantity divided by the number of days in the delivery month
Delivery period	The entire calendar month of the contract. Delivery begins on the first calendar day of the month and ends on the last day.
Last trading day	Three business days prior to the first day of the delivery month. Example: the March contract stops trading at the close on the third-to-last business day of February.
Trading hours	CME Globex: Sunday–Friday 18:00–17:00 ET (23-hour); liquid hours approximately 07:00–14:30 ET on business days
Open interest	Largest in months 1–6 and at the winter peaks (Nov–Mar); meaningful open interest through 36 months; activity thins but extends to 12 years for long-dated hedgers
Price limit	$3.00/MMBtu initial daily limit; expandable
Contract value at $3/MMBtu	$30,000 per contract

1.1 Henry Hub — The Delivery Point

Henry Hub is a physical natural gas distribution point operated by Sabine Pipe Line LLC in Erath, Louisiana. Its central position in the US pipeline network makes it the natural pricing reference for domestic natural gas:

- Connected to approximately 16 interstate and intrastate pipelines, including Southern Natural Gas, Transcontinental Gas Pipeline (Transco), Tennessee Gas Pipeline, Texas Eastern Transmission, Trunkline, Acadian, and others — providing access to supply from the Gulf of Mexico, Permian Basin, Haynesville Shale, and Appalachian Basin, and demand access to the US Southeast, Mid-Atlantic, New England, and Midwest markets.
- The Henry Hub spot price is the basis for a large proportion of physical gas contracts in North America. Regional prices at other hubs trade at a differential to Henry Hub (the "basis differential") that reflects local supply/demand balance, pipeline constraints, and transportation costs. Henry Hub NYMEX futures do not directly reflect regional pricing — physical traders hedge regional basis risk separately using basis swaps or regional futures products.
- Physical delivery at Henry Hub requires pipeline scheduling, nominations, and balancing — making the front-month contract converge to the physical spot market at Henry Hub in the final trading days.

The convergence of the front future to the physical spot is the mechanism that keeps the futures market anchored to physical market fundamentals.

1.2 Price Quotation and Spread Arithmetic

- Spreads are quoted as the price of the near month minus the price of the far month. A positive spread (contango) means the far month is more expensive — the market is paying for storage. A negative spread (backwardation) means the near month is more expensive — current demand exceeds available supply, or storage is being drawn down faster than expected.
- For natural gas, backwardation in the winter months is common and expected — winter demand is high, so winter contracts trade at a premium to adjacent months. Contango is typical across the shoulder seasons (spring and autumn) when storage injection occurs.
- DV01 of a single NG contract: a $0.01/MMBtu move = $100 per contract. A spread position (long one contract / short adjacent) has no outright price risk to parallel moves but full sensitivity to the slope between the two months.

2. The Storage Cycle — The Central Balancing Mechanism

Natural gas cannot be consumed instantaneously from the wellhead — the mismatch between production (relatively steady) and demand (highly seasonal and weather-driven) is bridged by underground storage. Storage levels are the most important fundamental variable for the shape of the NG forward curve, particularly the front 12 months.

2.1 US Natural Gas Storage Infrastructure

Storage Type	Working Capacity	Key Regions	Characteristics	Spread Implication
Depleted oil/gas reservoirs	~75% of total	Gulf Coast, Mid-Continent, Appalachia, Midwest	Largest category. Lower injection/withdrawal rates relative to capacity. Year-round cycling.	Baseline seasonal storage — drives the injection/withdrawal cycle that shapes the spring/autumn curve
Aquifer storage	~12% of total	Midwest, Mid-Atlantic	Require more cushion gas. Higher withdrawal flexibility than reservoirs in some cases.	Regional balancing; less directly relevant to Henry Hub spread structure
Salt cavern storage	~13% of total	Gulf Coast (primary), Midwest	Highest injection/withdrawal rates. Can cycle multiple times per year. Critical for peak-day/peak-week demand response.	Salt cavern availability drives the market's ability to respond to sudden demand spikes — key for front spread behaviour in cold events
LNG peaking facilities	Small	Northeast, Mid-Atlantic	Vaporise stored LNG for peak demand. Very high withdrawal rate for short periods.	Relevant for extreme cold events in the Northeast — reduces squeeze risk on January contracts

- **Total working gas capacity**: approximately 4.0–4.2 trillion cubic feet (Tcf). The US has never filled storage completely — the practical limit is approximately 3.9 Tcf (reached in October 2023). Storage filling beyond 90% of capacity begins to constrain the market's ability to accept injections, which pressures the front of the curve bearishly and compresses the summer injection-season spread (Mar–Oct).

2.2 The Injection and Withdrawal Seasons

The US natural gas storage cycle follows a well-defined annual pattern that is the foundation of the seasonal spread structure:

Season	Period	Direction	Typical Change	Curve Effect	Key Risk
Withdrawal	Nov–Mar	Storage drawn down by winter heating demand	−1.5 to −2.0 Tcf	Near months (Jan–Mar) trade at premium — backwardation. Front spreads tight or negative (nearby > deferred).	Cold weather outpacing withdrawal capacity or supply disruption creates extreme backwardation
Injection	Apr–Oct	Storage refilled from production surplus over demand	+2.0 to +2.5 Tcf	Deferred months trade at premium — contango. Spread (Mar–Apr "shoulder") prices the injection season carry.	Hot summer driving power demand; hurricane disrupting Gulf Coast supply; slow injection pace

Shoulder (spring)	Mar–Apr	Transition: withdrawal ends, injection begins	Variable	Mar–Apr spread is the most watched seasonal spread. Mar trades at seasonal withdrawal premium; Apr begins injection season pricing.	Late cold snap extending withdrawal; storage entering injection season at low levels
Shoulder (autumn)	Oct–Nov	Transition: injection ends, withdrawal begins	Variable	Oct–Nov spread marks the end of injection season. Storage level at this point determines winter pricing tightness.	Early cold snap pulling demand before storage is full; hurricane-reduced supply

2.3 EIA Weekly Storage Report — The Primary Weekly Signal

The EIA publishes weekly underground storage data every Thursday at 10:30 ET. This is the single most important scheduled weekly data event for the NG curve:

- **Working gas in storage (Bcf)**: the headline figure. Reported as total US and broken out by region: East, Midwest, Mountain, Pacific, and South Central (which includes the salt cavern facilities).
- **Week-on-week change (injection or withdrawal)**: compared to consensus analyst estimates, prior year, and the 5-year average. A storage report that shows a smaller-than-expected injection (or larger-than-expected withdrawal) is immediately bullish for the front contract and near-term spreads.
- **Storage vs 5-year average and vs prior year**: the percentage deviation from the 5-year average is the most widely watched context metric. Storage significantly below the 5-year average entering winter creates a tighter winter supply buffer and is bullish for November–March contracts. Storage significantly above the 5-year average entering injection season is bearish for near-term contracts.
- **The "injection season" fill target**: the market monitors whether the US is "on pace" to refill storage to a level sufficient to supply winter demand. A rule of thumb is that entering November with at least 3.5 Tcf provides adequate buffer for an average winter; entering with 3.2 Tcf or less in a context of cold weather forecasts creates acute near-month tightness.

The storage report surprise (actual vs consensus) is the primary driver of front NG spread volatility on a weekly basis. A 10 Bcf surprise (roughly 10% of average weekly injection/withdrawal) can move the front contract 5–15 cents/MMBtu within minutes of release.

3. Weather — The Dominant Demand Variable

Natural gas demand is more weather-sensitive than any other major commodity. A single week of extreme cold can increase US natural gas demand by 30–50 Bcf/day above a mild-weather baseline — a swing equivalent to 30–50% of total US production. Weather data, forecasts, and anomalies are therefore the most time-sensitive fundamental inputs for the NG curve.

3.1 Heating Degree Days (HDD) and Cooling Degree Days (CDD)

- **Heating Degree Days (HDD)**: calculated as max(0, 65°F − average daily temperature). Each HDD implies a baseline level of residential and commercial heating demand. HDD are reported weekly by NOAA and forecast by commercial weather services (DTN, WeatherBell, Maxar/Weather Trends International). The weekly EIA natural gas demand model is essentially a regression of gas consumption on population-weighted HDD.
- **Cooling Degree Days (CDD)**: calculated as max(0, average daily temperature − 65°F). CDDs drive summer electricity demand and therefore natural gas consumption by power generators. A hot summer increases gas burn by power plants significantly. CDDs are the primary weather variable for the summer (June–September) NG spread.
- **Population-weighted HDD/CDD**: national HDD figures are weighted by the population of each region to reflect the concentration of gas heating demand in the Midwest, Northeast, and Mid-Atlantic. A very cold week in the upper Midwest is more gas-impactful than the same temperature deviation in the Mountain states.

3.2 Weather Forecasting and the NG Market

The natural gas market tracks multiple weather forecast models and updates continuously as forecasts change:

- **European Centre for Medium-Range Weather Forecasts (ECMWF)**: generally considered the most accurate global weather model for 6–15 day forecasts. The ECMWF "Euro model" is the reference model for natural gas weather traders. A shift in the ECMWF ensemble toward colder or warmer forecasts for the next 1–2 weeks moves the front NG contract immediately. The 8–14 day and 6–10 day outlooks from ECMWF are the most watched.
- **GFS (Global Forecast System)**: the US NOAA model. Generally less accurate than ECMWF for 10–15 day outlooks but widely used and available free. When GFS and ECMWF diverge significantly, the market holds uncertainty premium until the models converge. Model "runs" update four times per day (00Z, 06Z, 12Z, 18Z).
- **CPC (Climate Prediction Center) 8–14 day and monthly outlooks**: NOAA's probabilistic temperature and precipitation outlooks for the medium and long range. Published weekly (8–14 day) and monthly. Provides the statistical framework for seasonal forecasting used by gas traders looking beyond the deterministic 10-day window.
- **ENSO and seasonal forecasts**: El Nino-Southern Oscillation affects winter temperature patterns in the US. El Nino winters in the northern US tend to be warmer than average (bearish for winter NG contracts). La Nina winters tend to be colder in the northern tier (bullish for winter contracts). These signals have a 3–6 month lead time and affect the pricing of the November–March strip in summer months.

3.3 The Gas-Weighted Degree Day (GWDD)

Commercial weather services produce "gas-weighted degree days" — HDD weighted by historical per-degree-day gas consumption by region. GWDDs are a more precise measure of gas demand impact than raw HDD because they account for the different gas intensity of heating per degree in each region:

- The Northeast (particularly New England and New York) has high per-HDD gas consumption because it lacks alternative heating infrastructure and has high population density. Cold in the Northeast has a disproportionate impact on national gas demand and on Algonquin/Transco pipeline basis.
- The South Central region has much lower per-HDD gas consumption for heating but high per-CDD gas consumption for power generation in summer.
- GWDD forecasts for the current week and the following 1–2 weeks are the most actionable weather variable for front-spread traders.

4. Supply Fundamentals

4.1 US Dry Natural Gas Production — The Shale Revolution

US dry natural gas production has grown from approximately 21 Bcf/day in 2008 to approximately 100–104 Bcf/day by 2023–2024, driven almost entirely by horizontal drilling and hydraulic fracturing in shale and tight-sand formations. This production growth transformed the US from a gas-importing country (with LNG import terminals under construction) to the world's largest natural gas producer and, since 2016, a growing net exporter. The supply landscape is now defined by the major producing basins:

Basin	Region	Production (approx.)	Primary Gas Type	Supply Characteristics	Key Data
Appalachian (Marcellus/Utica)	PA, WV, OH	33–35 Bcf/day	Dry gas (Marcellus dry), wet gas (Utica, SW Marcellus)	Largest single gas-producing basin globally. Very low breakeven (~$1.50–2.50/MMBtu). Takeaway capacity occasionally constrained in winter — key basis risk region.	EIA DPR, PA DEP, WV DEO, Enverus
Permian Basin	TX, NM	21–24 Bcf/day	Associated gas (produced with oil)	Gas production tied to oil drilling economics, not gas prices. Flaring historically high when takeaway limited. Growing rapidly with oil production.	EIA DPR, Texas RRC, operator guidance
Haynesville	LA, TX	14–17 Bcf/day	Dry gas	Located near Gulf Coast demand and LNG export terminals. Most price-responsive basin — producers ramp here when gas prices rise. Key swing supplier for LNG feed gas.	EIA DPR, Louisiana DNR, Enverus
Eagle Ford	South TX	6–7 Bcf/day	Wet gas / associated	Predominantly associated with oil. Proximity to Gulf Coast markets.	EIA DPR, Texas RRC
Anadarko / SCOOP/STACK	OK	5–6 Bcf/day	Dry and wet gas	Mature basin. Moderate growth potential.	EIA DPR, Oklahoma CC
DJ Basin / Piceance	CO, WY	3–4 Bcf/day	Associated and dry	Mostly associated with oil. Regional supply for Rockies/Mountain markets.	EIA DPR, Colorado OGCC

4.2 Associated Gas — The Oil Drilling Linkage

A significant and growing share of US natural gas production is "associated gas" — gas produced alongside crude oil, primarily in the Permian Basin. This creates a structural feature unique to natural gas: a portion of gas supply is determined by oil drilling economics, not gas prices:

- When oil prices are high and oil drilling activity increases, Permian associated gas production rises even if gas prices are low. This creates a "ratchet" effect: sustained high oil prices eventually increase gas supply and weigh on the front NG curve regardless of gas market fundamentals.
- The US oil rig count (Baker Hughes, weekly) is therefore a leading supply indicator for associated gas 3–6 months forward — the same way it is for oil supply, but with the opposite price incentive: low gas prices do not stop associated gas growth if oil prices remain supportive of drilling.
- Permian Basin gas flaring: when pipeline takeaway capacity is insufficient for the volume of associated gas produced, producers flare (burn) the excess. Texas Railroad Commission data on Permian flaring volumes provide a real-time measure of stranded gas that is not reaching the market. New pipeline completions that reduce flaring add supply.

4.3 Dry Gas Production Response to Price

- **Haynesville Shale as the swing basin**: Haynesville is the most price-responsive dry-gas basin. Producers in Haynesville (Comstock Resources, Southwestern Energy, Chesapeake/Expand Energy) have relatively short cycle times and proximity to Gulf Coast markets. When Henry Hub prices rise above approximately $3.00–3.50/MMBtu, Haynesville drilling activity increases. When prices fall below $2.00/MMBtu, drilling declines.
- **Appalachian takeaway constraints**: Marcellus/Utica gas is often cheaper to produce than Haynesville but faces periodic takeaway constraints in winter, when pipeline capacity to New England and Mid-Atlantic is fully utilised. This creates sharp regional basis widening (Algonquin Citygates vs Henry Hub) during cold snaps — a distinct spread opportunity in regional gas markets.
- **EIA Drilling Productivity Report (DPR)**: monthly publication tracking new-well gas production per rig and legacy decline by basin. The legacy decline in shale gas wells is steep — approximately 60–75% in the first year. This maintenance-drilling treadmill means gas production falls relatively quickly when rig counts drop.

4.4 Gulf of Mexico (GoM) Supply

- Offshore GoM natural gas production has declined sharply from its peak (~15 Bcf/day pre-Katrina) to approximately 2.0–2.5 Bcf/day currently. The decline reflects the depletion of legacy offshore gas fields and the economics of offshore gas vs onshore shale. GoM gas production is tied to oil-focused deepwater development and is no longer a primary supply variable for the NG curve.
- However, GoM supply is highly vulnerable to hurricane disruptions. A major hurricane passing through the GoM can shut in 0.5–2.0 Bcf/day of gas production for days to weeks, creating immediate front-spread bullish pressure.
- Hurricane season (June 1 – November 30) creates a recurring annual supply disruption risk that is priced into the summer and early autumn NG curve — particularly for July, August, September, and October contracts.

4.5 Canadian Pipeline Imports

- Canada supplies approximately 7–9 Bcf/day of natural gas to the US via pipeline, primarily from Alberta and British Columbia through the NOVA/TC Energy system. Canadian imports are most important for the US Pacific Northwest and Upper Midwest markets.
- Canadian supply has declined as a share of US supply due to growing US domestic production. Major Canadian pipeline constraints (TC Energy Coastal GasLink, NOVA Gas Transmission System capacity) occasionally affect import levels.

5. Demand Fundamentals

US natural gas demand totals approximately 85–95 Bcf/day averaged across the year, with peaks of 130–160 Bcf/day during extreme cold events and troughs of 60–70 Bcf/day during mild spring days. Demand is split across four primary sectors:

Demand Sector	Share of Total	Seasonal Pattern	Key Drivers	Curve Sensitivity
Residential & Commercial (heating/cooking)	~30%	Strongly winter-peaked. Peak Dec–Feb; trough May–Sep.	Temperature (HDD), population growth, building efficiency, gas vs heat pump switching	Primary driver of winter backwardation and summer contango. Front contract highly sensitive.
Electric Power Generation	~38%	Dual-peaked: winter (heating) and summer (A/C cooling demand). Spring and autumn troughs.	Temperature (CDD in summer), coal-to-gas switching economics, renewables output, carbon pricing	Most volatile sector. Determines mid-curve summer spread. Coal-to-gas switching creates a floor for gas demand at low prices.
Industrial	~23%	Relatively flat seasonally but cyclical with industrial output.	Manufacturing activity, fertiliser production, LNG plant fuel use, petrochemical feedstock	Slow-moving; affects back-curve baseline demand. Petrochemical expansion projects matter.
Pipeline & LNG Exports	~14–16% and growing	LNG exports relatively flat year-round (constrained by terminal capacity). Pipeline exports to Mexico vary seasonally.	LNG terminal utilisation, Mexico pipeline infrastructure, global LNG demand vs US gas price	Growing structural demand; LNG export growth shifts the entire curve higher. Most important back-curve demand driver since 2016.

5.1 Power Generation Demand — The Coal-to-Gas Switching Threshold

Power sector natural gas demand is the most dynamic demand segment and the primary mechanism through which the gas market self-corrects at low prices:

- **Coal-to-gas switching**: when natural gas prices fall below approximately $2.00–2.50/MMBtu (depending on coal prices and regional heat rates), gas becomes competitive with or cheaper than coal for power generation. At these prices, utility dispatchers switch from coal units to gas units, increasing gas demand materially. This creates a price floor in the front NG market during periods of gas oversupply.
- **Reverse switching (gas-to-coal) at high gas prices**: when gas rises above approximately $3.50–4.50/MMBtu, marginal coal units become competitive again (where coal capacity still exists). The remaining US coal fleet acts as a demand shock absorber at high gas prices.
- **Renewables intermittency and gas as backup**: wind and solar generation is variable. Gas peaker plants and combined-cycle units provide the dispatchable backup when renewable output falls. As the renewable share of the generation mix grows, the demand for gas as backup becomes more volatile — a calm, sunny period can sharply reduce gas burn; a cloudy, windless week can sharply increase it.
- **ERCOT (Texas) gas demand**: Texas is the largest gas-consuming state. The ERCOT grid is isolated from the national grid and relies heavily on gas generation. Extreme cold events in Texas (February 2021 Winter Storm Uri) create acute regional gas demand spikes that cascade through Henry Hub pricing.

5.2 LNG Exports — The Global Demand Connection

The development of US LNG export terminals since 2016 has fundamentally changed the demand structure of the US gas market. LNG exports connect the previously landlocked US market to global gas prices for the first time:

- **Operational LNG export terminals (as of 2024)**: Sabine Pass (Cheniere, ~30 Mtpa capacity), Corpus Christi (Cheniere, ~15 Mtpa), Freeport LNG (~15 Mtpa), Cameron LNG (~15 Mtpa), Sabine Pass Train 6 (2.0 Mtpa), Elba Island (Shell, ~2.5 Mtpa), and Venture Global Calcasieu Pass (~10 Mtpa). Total operational capacity approximately 14–15 Bcf/day of feed gas demand — roughly 14–15% of total US production.
- **LNG feed gas demand is relatively inelastic in the short run**: once committed to a train run, LNG plants operate continuously (capacity factor ~90%+). Planned and unplanned outages at LNG terminals (Freeport LNG explosion/shutdown in June 2022 removed ~2 Bcf/day of demand for several months) are immediate bearish events for front NG contracts.
- **LNG export-driven demand growth**: each Bcf/day of incremental LNG export capacity adds permanently to the baseline US gas demand — shifting the equilibrium storage balance and, over time, the level at which the forward curve settles. Approved but not-yet-built terminals (Plaquemines LNG, CP2, Port Arthur LNG) are back-curve bullish signals when they receive final investment decision (FID).
- **Global LNG price signal**: when international gas prices (TTF in Europe, JKM in Asia) are significantly above Henry Hub on a BTU-equivalent basis, the economics of US LNG exports are highly profitable. This incentivises maximum LNG plant utilisation and accelerates new terminal investment. When TTF/JKM are near Henry Hub parity, the marginal cargo may not be economic, reducing the demand pull on US gas.
- **The "LNG-linked" global gas market**: since 2022, US Henry Hub prices have shown increased correlation with TTF (the European gas benchmark) and JKM (Asian LNG prices) due to the growing volume of US LNG exports. A European energy crisis (Nord Stream disruption in 2022) pulled US LNG exports to Europe, reducing US storage injection pace and supporting Henry Hub — a demand shock transmission that would not have occurred before the LNG export terminal buildout.

5.3 Mexico Pipeline Exports

- US pipeline exports to Mexico have grown to approximately 6–7 Bcf/day, driven by Mexican power sector demand and the construction of cross-border pipelines (Sur de Texas-Tuxpan, Waha-to-Juarez, Wahalajara system). Mexico lacks domestic gas production growth and relies increasingly on US imports.
- Mexican demand is seasonal (higher summer air conditioning demand) and variable with Mexican economic conditions and power sector investment. Permian gas is the primary source for Mexico exports (geographic proximity via south Texas pipeline infrastructure).

6. The Forward Curve Structure — Zones and Drivers

The NG forward curve has a distinctive shape driven by the seasonality of demand and storage economics. Understanding how each zone of the curve is priced is the prerequisite for spread and fly positioning.

Zone	Months (approx.)	Primary Driver	Secondary Driver	Typical Shape	Key Signal
Prompt	C1–C2	Current storage vs 5-yr avg; weather forecast (next 7–14 days)	Production vs pipeline nominations	Highly variable — backwardation in winter; contango in spring/autumn	EIA weekly storage; ECMWF/GFS 10-day temperature outlook
Near-term	C2–C5	Seasonal demand forecast; storage trajectory for season end	Hurricane risk (summer); weather outlook beyond 14 days	Injection season: mild contango. Withdrawal season: backwardation building toward winter peak.	CPC 8–14 day outlook; ENSO forecast for seasonal signal; EIA STEO supply/demand
Winter strip	Nov–Mar	Storage entering withdrawal season; ENSO/winter weather probability	LNG export demand; production trajectory	Winter contracts typically trade at a premium to adjacent spring/autumn months. Size of premium = market's assessment of winter tightness.	October end-of-injection-season storage level; ENSO update; winter weather outlook
Summer strip	Apr–Oct	Storage injection economics; power sector demand (CDDs)	LNG feed gas demand; coal-to-gas switching level	Mild contango reflecting storage carry cost. Steeper if storage starts injection season low.	EIA STEO power burn forecast; coal retirements; LNG terminal utilisation
Mid-curve	C12–C24	LNG export growth trajectory; long-run supply investment; r* for gas	Energy transition pace; renewable buildout	Reflects structural supply/demand balance. LNG FIDs shift this zone.	LNG terminal FID announcements; EIA Annual Energy Outlook; Permian/Haynesville rig count trend
Back curve	C24+	Energy transition (gas demand peak timing); structural LNG demand; carbon pricing	Long-run producer breakeven; domestic demand destruction from electrification	Anchored by long-run marginal cost of supply. Moves slowly.	EIA/IEA long-term outlooks; US power sector electrification trajectory; carbon price signals

6.1 The Winter Premium — The Most Important Structural Feature

In most years, November, December, January, and February contracts trade at a significant premium to the adjacent months. This winter premium reflects:

- **Peak demand concentration**: residential and commercial heating demand is concentrated in a 4–5 month window. The market must price a premium sufficient to incentivise producers and storage operators to accumulate supply through spring and summer to meet this concentrated demand.
- **Storage drawdown premium**: the ability to withdraw gas from storage during winter is a service that has value beyond the gas commodity itself. Storage operators earn this premium by holding gas through injection season and selling it in winter — the "winter/summer spread" trade.
- **Supply disruption risk premium**: winter production can be impaired by wellhead freeze-offs (common in the Permian Basin and Appalachia during extreme cold). The market prices a risk premium for this potential supply disruption in the winter months, above and beyond what storage levels imply.

- **The March/April "shoulder spread"**: the spread between March (last withdrawal month) and April (first injection month) is one of the most watched spreads in the NG market. In tight winters, March trades at a large premium to April as storage is drawn down and supply tightens. In mild or oversupplied winters, March trades near or below April.

6.2 The Summer-Winter Spread (Calendar Spread)

The spread between a summer contract (e.g. July) and an adjacent winter contract (e.g. January of the following year) is the fundamental "injection season trade" — storing gas in summer to sell in winter:

- The economic cost of storing gas from July to January includes: physical storage cost (~$0.10–0.25/MMBtu/month), financing cost (rate-dependent), and basis risk (Henry Hub prices at delivery vs entry). The July-January spread must exceed these costs to make the storage trade profitable.
- When the July-January spread is wide (winter contract significantly above summer), storage operators inject aggressively, building inventory and placing downward pressure on the winter premium as storage fills. This self-correcting mechanism compresses the spread over time unless demand is structurally stronger than supply.
- When the spread is narrow or inverted (summer contract at or above winter), the storage trade is uneconomic. Injection slows, and if sustained, the market enters winter with lower-than-average inventory — which then reprices the winter premium higher.

7. Documented Seasonal Patterns

Natural gas has the most pronounced and consistent seasonal patterns of any major commodity futures market, driven by the regularity of the heating and cooling demand cycles and the storage injection/withdrawal calendar. These patterns are reliable baselines but can be substantially overridden by weather anomalies, supply disruptions, and LNG demand shifts.

Period	Storage Action	Weather Driver	Typical Spread Behaviour	Key Risk / Override
Jan–Feb (peak winter)	Maximum withdrawal rates. Storage typically below 2.5 Tcf.	HDD — coldest weeks of year. Polar vortex events create extreme demand spikes.	Near-month backwardation at maximum. C1 at large premium to C2–C3. January contract most volatile.	Polar vortex event can push front spreads to extreme backwardation ($0.50–$2.00/MMBtu); wellhead freeze-offs add supply shock.
Mar (late winter/transition)	Withdrawal continues but decelerates. Injection season approaches.	HDD declining but late cold snaps frequent in North. High uncertainty.	March often at premium to April (shoulder spread). The size of this spread is a key market signal about winter balance.	Late cold snap in March (2019, 2022) pushes March into extreme premium and extends backwardation.
Apr–May (spring injection begins)	Injection season begins. Storage rebuilding.	Mild temperatures. CDD minimal. Weather impact lower.	Contango building as summer storage economics dominate. April-May spread mildly positive (contango).	Slow injection pace (production disruption or warm spring increasing power demand) can steepen contango pressure on summer months.
Jun–Aug (summer, power burn peak)	Active injection. Storage typically building toward 3.0–3.5 Tcf.	CDD — air conditioning demand. Hot summers increase power burn materially.	July and August at modest premium to adjacent months reflecting power burn demand. Hurricane season adds risk premium.	Heatwave (2023 record heat) dramatically increases power burn; hurricane shutting GoM supply. Both are bullish for summer contracts.
Sep–Oct (end of injection)	Injection decelerating. Storage near seasonal maximum. Oct 31 "end-of-injection" level is the key metric.	CDD declining. HDD not yet significant. Weather impact lowest.	Nov-Oct spread (first winter spread) begins to price winter tightness or comfort. Wide Oct-Nov spread = market expecting tight winter.	Storage entering October below 3.4 Tcf in context of cold winter forecast creates early bullish repricing of winter strip.
Nov–Dec (early winter)	Withdrawal season begins. Storage direction turns.	HDD beginning to accumulate. First major cold snaps of season.	November and December at significant premium to following spring (April). Winter strip premium builds with cold forecasts.	ENSO flip (El Nino to La Nina) or early polar vortex forecast shift creates sharp winter strip repricing.

7.1 The "Widow Maker" — The March-April Spread

The March-April calendar spread in natural gas is informally known as the "widow maker" due to its history of extreme and unpredictable moves. It is the single most volatile spread in the NG market:

- March is the last month of the withdrawal season; April is the first month of the injection season. This structural transition creates a potential cliff between the winter premium (March) and summer pricing (April).
- In years of extreme cold or supply disruption, March can trade $1.00–$5.00/MMBtu above April as storage approaches critically low levels and the market prices a genuine physical shortage risk.

- In mild winters or oversupplied markets, March may trade near par with or even below April as the winter premium evaporates.
- The spread is driven by: storage levels entering March, weather forecasts for the remaining heating season, production rates, and the pace of LNG exports.

Traders who are long the March-April spread entering winter face the risk of a mild end-of-winter that collapses the premium dramatically in a short period. Traders who are short face the risk of a cold snap that sends the spread to multi-dollar levels. Position sizing and stop-loss discipline are critical.

8. Infrastructure — Pipelines, LNG Terminals, and Constraints

Natural gas is a pipeline commodity. Unlike crude oil, it cannot be easily stored in tankers or moved by alternative transport modes (LNG is an exception but requires specialised facilities). Pipeline infrastructure — its capacity, constraints, and expansions — directly affects regional supply/demand balance and the basis differentials between regional hubs and Henry Hub.

8.1 Key Interstate Pipeline Systems

Pipeline	Route / Direction	Capacity (approx.)	Market Significance
Transcontinental Gas Pipeline (Transco — Williams)	Gulf Coast to New York/New England via Appalachia	~20 Bcf/day	The largest US natural gas pipeline by volume. Serves the most weather-sensitive demand region (Northeast). Transco Zone 6 NY basis is the most watched regional basis in the US gas market. Winter constraints on Transco create the Algonquin/Transco basis spikes during cold events.
Southern Natural Gas (Soco — Berkshire Hathaway)	Gulf Coast to Southeast / Appalachian markets	~8 Bcf/day	Serves the Southeast demand corridor and connects to Appalachian supply. Secondary to Transco but important for Mid-Atlantic.
Tennessee Gas Pipeline (Algonquin — Enbridge / TC Energy)	Gulf Coast to New England via Midwest	~10 Bcf/day	Second major Northeast corridor. Algonquin Citygates (AGT) basis is the benchmark for New England gas pricing. Extreme winter basis spikes at AGT (seen at $100+/MMBtu during Polar Vortex 2014) reflect pipeline capacity constraints.
Texas Eastern Transmission (TETCO — Enbridge)	Gulf Coast to Appalachia and Northeast	~12 Bcf/day	Major Gulf-to-Northeast artery. Connects Gulf Coast production to Eastern demand. TETCO M-3 is an important Mid-Atlantic basis point.
Rockies Express Pipeline (REX — Tallgrass)	Rockies to Appalachia (bidirectional)	~4 Bcf/day	Reversed eastbound to move Appalachian gas westward when Marcellus was oversupplied. A key pipeline for balancing Appalachian/Rockies supply.
TC Energy NOVA / GTN	Canadian BC/Alberta to Pacific Northwest and US Midwest	~10–15 Bcf/day	Primary vehicle for Canadian gas imports into the US. TC Energy outages (NOVA maintenance) create temporary reductions in US Canadian imports.

8.2 Regional Basis Markets

Henry Hub NYMEX futures price gas at one specific physical location. Gas consumed or produced in other regions trades at a differential (basis) to Henry Hub that reflects pipeline economics, local supply/demand, and seasonal patterns:

- **Algonquin Citygates (AGT)** — New England. Historically the most volatile basis in the US. Extreme winter cold events cause AGT to trade $5–$70/MMBtu above Henry Hub (vs typical $0.20–0.50 premium) when pipeline capacity to New England is fully utilised and no additional gas can physically reach the region.
- **Transco Zone 6 NY** — New York / Mid-Atlantic. The benchmark for the Northeast's largest gas market. Correlated with AGT but typically less extreme. Basis spikes are common on cold days.
- **Waha Hub (West Texas / Permian)** — the delivery point for Permian Basin gas. When Permian associated gas production exceeds pipeline takeaway capacity, Waha basis goes deeply negative (Waha has traded at −$8 to −$10/MMBtu vs Henry Hub during periods of severe pipeline

congestion). Waha basis is an important signal for Permian gas flaring levels and takeaway capacity utilisation.

- **Chicago Citygates** — Midwest demand centre. Reflects Appalachian and Canadian supply access to the largest Midwest demand market. Typically small premium or at par with Henry Hub.
- **Dominion South (Appalachia)** — the primary Marcellus/Utica producing region basis. Historically traded at a significant discount to Henry Hub during Appalachian takeaway constraints (2012–2016). Improved as Leach Xpress, Atlantic Sunrise, and other pipelines expanded.

9. Macro and Cross-Market Drivers

9.1 The Gas-Coal Switching Relationship

As described in Section 5, gas-coal switching in the power sector creates a demand-side price floor and ceiling for natural gas that is directly observable and quantifiable:

- **The switching price range**: approximately $2.00–2.50/MMBtu Henry Hub for coal-to-gas switching (gas becomes cheaper than coal per MWh generated at most US coal plants). Approximately $3.50–5.00/MMBtu for gas-to-coal switching (residual coal plants become competitive). These thresholds move with coal prices and regional heat rates.
- **CAPP vs NAPP coal prices**: Central Appalachian (CAPP) and Northern Appalachian (NAPP) coal prices (available from S&P; Global Platts, Argus) determine the coal-side of the switching calculation. When coal prices rise, the switching threshold moves lower (gas becomes competitive vs coal at a lower gas price).
- **Power sector gas burn data (EIA)**: weekly and monthly EIA data on gas consumed for power generation provides a real-time read on the current level of coal-to-gas switching. Year-on-year changes in power burn are an important mid-curve demand signal.

9.2 The TTF-Henry Hub Relationship

The emergence of US LNG exports has created a new cross-market relationship between Henry Hub (domestic US gas) and TTF (Dutch Title Transfer Facility — the European gas benchmark):

- **TTF-Henry Hub spread**: when TTF trades significantly above Henry Hub (on an energy-equivalent basis, adjusting for liquefaction costs of approximately $2.50–3.50/MMBtu and shipping of $0.50–1.00/MMBtu), US LNG exports are economically attractive, pulling gas away from the US market and tightening domestic storage. This is a mid-curve bullish signal for NG.
- **LNG cargo diversions**: spot LNG cargoes can be redirected between Europe and Asia based on the TTF-JKM spread. High Asian demand (cold winter, supply disruption) pulls LNG away from Europe, tightening TTF, which then increases demand for US LNG — a transmission chain that ultimately affects Henry Hub storage.
- **The 2022 European energy crisis**: Russian natural gas supply reductions to Europe drove TTF to record levels (€300+/MWh). This pulled US LNG exports to near-maximum capacity, reduced US storage injection pace, and contributed to the 2022 Henry Hub rally. This demonstrated the now-established channel between global events and the previously isolated US market.

9.3 Crude Oil Price Relationship

- **Associated gas linkage**: as described in Section 4, Permian Basin associated gas supply grows with oil drilling. High oil prices that incentivise more Permian oil drilling eventually increase gas supply — a lagged bearish signal for NG.
- **Energy substitution at industrial level**: in certain industrial processes, gas and oil products (fuel oil, naphtha) are substitutable. High oil prices can redirect industrial demand toward gas, supporting gas prices at the margin.
- **No direct financial correlation with crude oil**: Henry Hub natural gas is not directly correlated with Brent or WTI in normal market conditions. The two markets are driven by different supply and demand fundamentals. Any correlation is mediated through the specific channels described above (associated gas, LNG economics, energy substitution) rather than a direct commodity relationship.

9.4 Speculative Positioning — COT Data

- **CFTC Commitments of Traders (COT)**: weekly NYMEX NG positioning data, published Fridays. Managed money net position is the primary sentiment indicator. Natural gas managed money positioning is highly seasonal: funds tend to build long positions ahead of winter (September–November) and reduce them as winter progresses.

- **Producer hedging**: natural gas producers (E&P; companies) hedge forward production by selling NYMEX NG futures or OTC swaps. Heavy producer hedging in the near months during high-price periods creates known selling pressure. The 13-F filings of large gas producers (quarterly) reveal the scale of their hedging programmes.
- **Extreme positioning as a contrary indicator**: when managed money net short in NG is at extreme levels (as it was in summer 2023 before the winter 2023-24 rally), the risk of a short-covering move on a weather catalyst is elevated. The reverse applies to extreme net longs.

10. Documented Supply Disruptions and Tail Events

Natural gas is uniquely vulnerable to acute supply and demand shocks because pipeline infrastructure has no substitutes in the short run. The following documented events illustrate the types of disruptions that can create extreme spread moves:

Event	Date	Type	Supply/Demand Impact	Front Spread Impact
Hurricanes Katrina & Rita	Aug–Sep 2005	GoM supply disruption	~15 Bcf/day shut-in; lasting infrastructure damage	Henry Hub spot exceeded \$15/MMBtu. Front contracts in extreme backwardation. Largest single supply shock in modern US gas market history.
Polar Vortex	Jan 2014	Extreme cold demand spike	National gas demand 100+ Bcf/day for multiple days; wellhead freeze-offs in Appalachia removed ~4–6 Bcf/day supply	Henry Hub spot exceeded \$20/MMBtu briefly. Algonquin basis >\$70/MMBtu. Front-month backwardation extreme.
Freeport LNG explosion	Jun 2022	LNG export terminal outage	~2 Bcf/day of LNG feed gas demand removed for ~6 months	Bearish for front NG — demand destruction. Henry Hub fell from ~\$8–9 to ~\$5/MMBtu over following weeks.
Winter Storm Uri (Texas)	Feb 2021	Combined supply & demand shock	~15 Bcf/day of supply lost (wellhead freeze-offs, processing failures); simultaneous record Texas heating demand	West Texas Waha spot >\$400/MMBtu briefly. Henry Hub spot ~\$23/MMBtu. ERCOT electricity crisis. Extreme regional basis dislocation.
Russia-Ukraine / Nord Stream	2022	Global LNG demand surge	European gas crisis pulled US LNG to maximum capacity, reduced US injection pace	Henry Hub rallied from ~\$4 to ~\$9/MMBtu 2022. Mid-curve pricing shifted structurally higher as LNG export volumes rose.
Record production 2023	2023–2024	Structural oversupply	US dry gas production reached 100+ Bcf/day, exceeding LNG + domestic demand	Henry Hub fell to multi-year lows (\$1.50–2.00/MMBtu). Front spreads in mild contango. Storage filled to near-record levels.

10.1 Wellhead Freeze-Offs — The Winter Supply Risk

- When temperatures fall far below freezing at producing wellheads (particularly in the Permian Basin and parts of Appalachia, which lack the Arctic-grade insulation used in Canadian production), water and condensate in the wellbore and gathering lines freeze, shutting in production. Freeze-offs can remove 5–15 Bcf/day of supply within 24–48 hours during a severe cold event.
- This creates the perverse outcome that the coldest weather events — which create the highest demand — simultaneously reduce supply. The combination is the most bullish scenario for the front NG contract and creates the most extreme backwardation in the prompt spread.
- Since Winter Storm Uri (2021), Texas has implemented some weatherisation requirements for wellheads and processing facilities, but the extent of compliance and the severity of future events remains an annual risk variable.

11. Key Data Sources and Release Schedule

Report	Publisher	Frequency	Release Timing	Primary Curve Impact
EIA Weekly Natural Gas Storage Report	US EIA	Weekly	Thu 10:30 ET	Prompt C1–C4 — the dominant weekly event
ECMWF 10-day temperature forecast update	ECMWF	4× daily	00Z/06Z/12Z/18Z	C1–C3 — most important weather model for NG traders
GFS model update	NOAA	4× daily	00Z/06Z/12Z/18Z	C1–C3 — secondary; watched when diverging from ECMWF
CPC 8–14 day temperature outlook	NOAA CPC	Weekly	Mon, ~11:00 ET	C2–C4 — medium-range weather signal
CPC monthly temperature outlook	NOAA CPC	Monthly	~15th of month	C3–C5 — seasonal demand outlook
ENSO forecast update	NOAA CPC	Monthly	~Mid-month	C4–C7 (winter strip) — seasonal temperature probability
EIA Short-Term Energy Outlook	US EIA	Monthly	2nd Tue of month	C2–C12 — supply/demand balance forecast; storage trajectory
EIA Drilling Productivity Report	US EIA	Monthly	~2nd Mon of month	C3–C12 — shale production and DUC count; supply outlook
Baker Hughes Rig Count	Baker Hughes	Weekly	Fri ~13:00 ET	C3–C12 — leading indicator for gas production 3–6 months
EIA Monthly Natural Gas Report	US EIA	Monthly	~4th week	C3–C6 — comprehensive supply/demand/storage data
EIA Annual Energy Outlook	US EIA	Annual	January–February	C12–C60 — long-run supply/demand trajectory; LNG exports
LNG export terminal utilisation	EIA / Kpler	Weekly	Various	C1–C6 — feed gas demand signal; terminal outage detection
CFTC Commitments of Traders	CFTC	Weekly	Fri ~15:30 ET	Positioning / crowding — all contracts
NatGasWeather / DTN / Maxar	Private	Daily	Continuous	C1–C3 — commercial weather services used by gas traders
Texas Railroad Commission flaring data	Texas RRC	Monthly	~2 month lag	C3–C6 — Permian associated gas flaring vs takeaway capacity
NOAA Atlantic Hurricane Forecasts	NOAA NHC	Seasonal + active season daily	Jun–Nov	C3–C5 (summer/autumn) — GoM supply disruption risk

NYMEX Natural Gas is the most weather-sensitive futures market covered in this series. No other major commodity can swing 30–50% in demand within a single week based on temperature alone. This makes the NG curve simultaneously the most seasonally predictable (the structural shape — winter premium, injection contango — repeats every year) and the most event-driven (a polar vortex, a hurricane, or a cold snap can compress months of fundamental positioning into a single week's price move). Storage levels relative to the 5-year average provide the structural context; weather forecasts determine the near-term direction; production and LNG export trajectories define the medium-term balance.

For spread traders, the most persistent edge comes from correctly reading the interaction between storage trajectory and seasonal demand: when storage enters winter below the 5-year average in the context of a cold ENSO forecast, the winter strip is structurally underpriced relative to historical norms. When storage is full and LNG is disrupted, the injection-season contango is insufficient to cover storage costs and will compress. The March-April spread is the single most concentrated expression of the seasonal balance — and its behaviour across the injection/withdrawal transition is the market's clearest statement about whether supply and demand are in equilibrium.